

LESSON 8-5 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Graphing Inequalities**

Maryland Standard 6.EE.B.8 • Learning target – I can graph an inequality on a number line using an open or a closed dot.

REMEMBER

- Use an **open dot** for $<$ or $>$. Use a **closed dot** for $\leq$ or $\geq$.
- Draw the arrow toward the solutions. The dot depends on the **symbol**, not on the sign of the number.
- Test a number under your arrow. It should make the inequality true.

WORKED EXAMPLEGraph $y > -2$.**Step 1** $>$ does not include -2 , so draw an open dot at -2 .**Step 2** Solutions are greater, so the arrow points right.**Answer** Open dot at -2 , arrow right**WORD BANK****open dot** – the number is NOT a solution ($<$ or $>$)**closed dot** – the number IS a solution ($\leq$ or $\geq$)**at least** – that number or more ($\geq$)**arrow** – points to all the numbers that work**PRACTICE***use the number lines • show your work*

1. For each graph, tell the dot (open or closed) and the arrow direction (left or right).

Step 1 Look at the symbol: $<$ or $>$ means open; $\leq$ or $\geq$ means closed.**Step 2** Greater ($>$ or $\geq$) points right. Less ($<$ or $\leq$) points left.

a. $x \geq 9$ _____

b. $n < -4$ _____

c. $p \leq 0$ _____

d. $k > 2.5$ _____

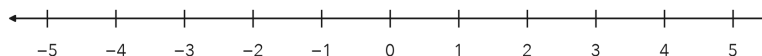
e. $w \geq -6$ _____

f. $r < 7$ _____

g. $b > -1.5$ _____

h. $c \leq 12$ _____

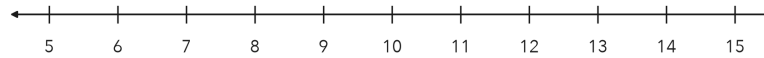
2. Graph $a \leq -1$ on the number line. Then name one solution.

Step 1 Find -1 on the number line.**Step 2** $\leq$ includes -1 , so choose the dot.**Step 3** Shade toward the numbers that are less.**One solution** _____

- 3. Hiking Trail.** Hikers must be **at least** 10 years old. Write an inequality for a , a hiker's age. Then graph it.

Step 1 "At least 10" means 10 or more.

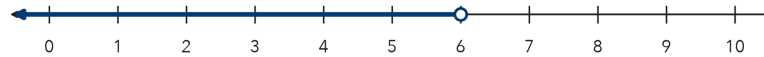
Step 2 Draw the dot at 10, then the arrow.



Inequality _____

- 4. Read the Graph.** Write the inequality shown, using m . Then describe a real situation it could show.

Frame The graph shows m ____ 6. A real situation is ____.



CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Ask someone at home for a rule with a limit (bedtime, screen time, a speed limit). Write it as an inequality and graph it.

LESSON 8-5 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Testing Solutions to Inequalities**

Maryland Standard 6.EE.B.8 • Learning target – I can test values against an inequality and graph the solution set.

REMEMBER

- **Substitute** the value. A **true** statement means it is a solution.
- An inequality usually has **many** solutions.
- The **boundary** is a solution for $\leq$ and $\geq$ only.

WORKED EXAMPLEWhich of 6, 8, and 11 are solutions of $x \geq 8$?**Step 1** Test 6: $6 \geq 8$ is false. Test 8: $8 \geq 8$ is true.**Step 2** Test 11: $11 \geq 8$ is true.**Answer 8 and 11 are solutions; 6 is not.****WORD BANK****substitute** – put the number in place of the letter**solution** – a number that makes it true**boundary** – the number in the inequality**true** – the comparison is correct**PRACTICE***show your work • label units*

1. Substitute each value. Write
- yes**
- if it is a solution or
- no**
- if it is not.

Step 1 Put the value in place of x .**Step 2** Read the comparison. True means yes; false means no.

Inequality	$x = 4$	$x = 7$	$x = 10$
$x > 7$			
$x \leq 7$			
$x \geq 4$			

2. Is the value a solution? Write
- yes**
- or
- no**
- .

a. $x < 9$; $x = 9$ ____

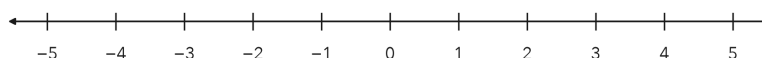
b. $m \geq -1$; $m = 0$ ____

c. $t \leq 2.5$; $t = 3$ ____

- 3.
- Bowling.**
- A bowling team wins a trophy if it scores
- more than**
- 450 points. Write an inequality for the score
- s
- . Test 450, 445, and 472.

Step 1 "More than" means $>$.**Step 2** Put each score in place of s . Is it true?**Inequality and winning scores** _____

4. Test
- -3
- ,
- 0
- , and
- 2
- in
- $y \leq -1$
- . Then graph
- $y \leq -1$
- on the number line.

Step 1 Put -3 , 0 , and 2 in place of y , one at a time.**Step 2** $\leq$ includes -1 , so choose the dot. Then draw the arrow.**Solutions from the list** _____

5. **Find the Error.** Bianca says $n < 12$ has 12 solutions: 11, 10, 9, and so on down to 0. Explain what she is missing.

Frame Bianca is missing _____. For example, _____ is also a solution.

TEST PRACTICE

choose the best answer

6. Which value is **not** a solution of $k \geq -2$?

A -3

B -2

C 0

D 5

7. Which value is a solution of $n > 15$?

A 12

B 14.9

C 15

D 15.5

CHECK YOURSELF

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